

Fernando Agustin Teodoro

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SUMMARY

Software developer who builds the plumbing between systems: data pipelines, signed API integrations, real-time feeds and the databases behind them. I've ingested public transit data on a nightly cloud schedule, joined messy open datasets, and built concurrency-safe booking in Postgres. I test against real data, deploy with Docker and Google Cloud, and review every AI-assisted change myself.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C, Dart

APIs & integration: REST (Django, Flask), HMAC-SHA1 signing, GTFS Realtime, Twilio, UDP, WebSockets

Data: PostgreSQL/PostGIS, MongoDB, SQLite, Pandas; schema design, row-level locking and security

Pipelines & cloud: Docker, Google Cloud (Run, Build, Scheduler), GitHub Actions, BeautifulSoup scraping

Configuration: YAML (GitHub Actions, cloudbuild.yaml, CMS config and front matter, pubspec), JSON

Testing: Mocha/Chai, unittest, node:test, Django tests; integration tests and recorded-data replays

AI-assisted development: Claude Code, GitHub Copilot; I own the plans and review every change

SOFTWARE PROJECTS

Melbourne Public Transport Widget

Mar – Dec 2025

Python (Flask) · MongoDB · SQLite · GTFS Realtime · Docker · Google Cloud (Run, Build, Scheduler) · GitHub Actions

- Implemented PTV's HMAC-SHA1 request signing and merged its API with GTFS Realtime protobuf feeds
- Built a Flask scraper that checks a second CKAN endpoint so the ~130 MB rebuild runs only on new data
- Versioned each ingested record and purged superseded data when a new release landed
- Deployed the scraper to Cloud Run via Cloud Build; Cloud Scheduler runs it at 3am, after PTV's refresh
- Added a GitHub Actions check that fails CI if a debug or mock flag is left on before merging
- Designed a 14-table SQLite cache with a tested merge helper, so the APIs are called only on a miss

Pa-Ride (Carpool Marketplace)

Aug 2026 – Present

Supabase (PostgreSQL, PostGIS, Auth) · TypeScript · React Native · Claude Code

- Designed a 26-migration Postgres schema with row-level security and explicit grants on every table
- Built seat booking as row-locked Postgres functions with server-derived availability, so seats can't be oversold even by a compromised client
- Built PostGIS corridor matching for drivers passing near a rider's route, with a detour cost and price
- Joined open Philippine place datasets for search, resolving renamed provinces and duplicate city names

C-LARA Image Annotation Tool (University Capstone)

Aug – Nov 2023

Python · Django REST Framework · React · Agile/Scrum · Product Owner, 5-person team

- Built the Django REST API (11 endpoints over images, words and coordinates) and its 37 tests; graded H1
- When the client's spec was ambiguous, shipped a simpler schema with a documented migration path
- Wrote handover docs and an integration report proposing API and schema extensions for C-LARA

Realtime Queue Management System

Jan 2026 – Present

Node.js (Meteor) · MongoDB · TypeScript · Mocha/Chai · Docker · Cloud Run · Cloud Build · GitHub Actions · Twilio

- Moved 27 methods to a thin-method standard: validate, check auth, hand off to a unit-testable function
- Traced a slow, staggered list load to an N+1 pattern (about three subscriptions per row); replaced it with one batched subscription plus indexes where loads had been full collection scans
- Made the server the only clock, so anonymous clients can't set timestamps to jump the queue
- Enforced each customer's feature access with server-side checks rather than hidden menus
- Built configurable patient notifications and tested them end to end over Twilio WhatsApp
- Grew the suite to 900+ unit and integration tests, run in GitHub Actions on pull requests into the release branch; each release auto-deploys to Cloud Run

Sim Racing Telemetry Dashboard (Gran Turismo 7)

Sep 2026

Python (standard library) · UDP · WebSockets · JavaScript · unittest / node:test · Claude Code

- Built a dependency-free Python bridge that decrypts PS4 UDP telemetry and relays it over WebSockets
- Traced lap times drifting from the game's own to frames timestamped on arrival over jittery Wi-Fi; switching to the game's clock cut lap-time error from 384 ms to 9 ms
- Wrote 166 tests, including replays of real PS4 recordings and an end-to-end UDP/WebSocket test

COVID-19 Air Quality Analysis

2021

Python · Pandas · NumPy · scikit-learn · 3-person university team · built without AI

- Built a Pandas pipeline joining four public datasets, back-filling missing postcodes by hand
- Chose Mutual Information over linear methods; found no significant link between mobility and air quality

Portfolio Site

Aug 2026 – Present

Python · Markdown/YAML · Pages CMS · GitHub Actions

- Built the Lab as a home for live demos, with each tool kept in its own repo and pulled in at build time

WORK EXPERIENCE

Freelance QA Tester | UK ETA Visa Services (iOS), via Upwork

May 2026

- Tested the full visa flow end to end instead of the brief's 10 to 15 minute pass, tracing how inputs affected later screens; earned a 5-star review

Ad-hoc Programmer | NutTAG Australia

Dec 2023 – Jan 2024

- Got up to speed on a product NutTAG had just inherited, helping the team understand it
- Redesigned the support chatbot's flow and knowledge base to target complaints in NutTAG's reviews

COMMUNITY

Australian Computer Society | member and event volunteer, 30+ CPD hours logged in 2026 2025 – Present

Google Developer Group Melbourne | attendee 2024 – Present

EDUCATION

BSc Computing & Software Systems | University of Melbourne

2023

- Coursework: Foundations of Algorithms (C), Computer Systems (C, socket programming), Object-Oriented Software Development (Java), Software Modelling and Design (Java)

International Baccalaureate Diploma | German-European School Manila

2018